FitBuddy: AI Fitness Plan Generator using Gemini Models
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Project Design Phase – FitBuddy

The **Project Design Phase** focuses on designing the overall structure, architecture, user interface, AI integration, and database components of the FitBuddy application.

1. System Architecture Design

FitBuddy follows a **modular FastAPI-based architecture** consisting of:

- **Frontend:** HTML + Jinja2 templates
- **Backend:** FastAPI
- **AI Layer:** Google Gemini Pro and Gemini Flash
- **Database:** SQLite with SQLAlchemy ORM

2. AI Model Design

- **Gemini 1.5 Pro** is designed for generating structured 7-day workout plans and updating plans based on feedback.
- **Gemini Flash** is designed for generating quick nutrition and recovery tips.

3. Database Design

The system stores:

- User ID
- Name
- Age
- Weight
- Fitness goal
- Workout intensity
- Original workout plan
- Updated workout plan

SQLite with SQLAlchemy ORM is used for persistent storage.

4. Backend Design

The main routes are designed as:

- `/` – Displays the user input page.
- `/generate-workout` – Generates the personalized workout plan and nutrition tip.
- `/submit-feedback` – Updates the workout plan using user feedback.
- `/view-all-users` – Provides the admin view of users and their plans.

5. User Interface Design

Three main Jinja2 templates are designed:

- **index.html** – User input form.
- **result.html** – Displays the personalized workout plan, nutrition tip, and feedback form.
- **all_users.html** – Admin panel for viewing user information and plans.

6. Workflow Design

The planned workflow is:

User Input → FastAPI Backend → Gemini AI → Personalized Workout/Nutrition Plan → Database Storage → User Feedback → Updated Plan

This design provides a structured flow between the frontend, backend, AI models, and database.

7. Final Design

The project is designed as a **user-friendly AI-powered fitness application** that combines personalized workout generation, nutrition guidance, feedback-based plan modification, and administrative monitoring.

Develop the FitBuddy design further

- Design the FitBuddy database schema
- Create the FitBuddy system architecture